Metabolite profiling of multiple measures of **adiposity**: A **Mendelian randomization** analysis

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Different **measures of increased adiposity** show similar **metabolic profiles** – can we use profiles to explore relationships with disease?

INTRODUCTION

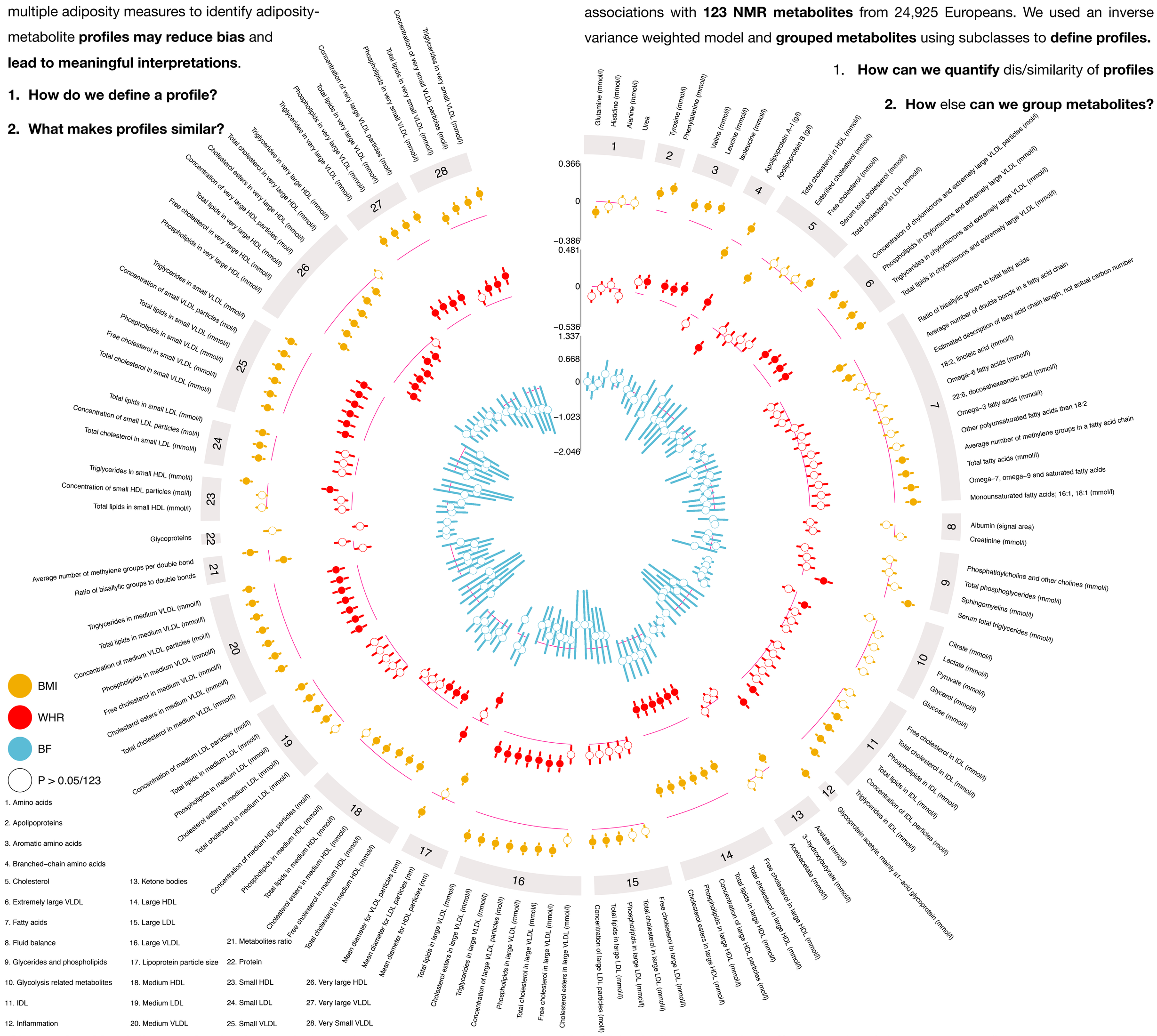
Adiposity is **associated with** many **diseases** and can be **measured in different ways**, each with its own individual nuances. **Metabolites** are part of the adiposity-disease biological pathway. **Identifying meaningful individual adiposity-metabolite associations is complicated** and **may be biased.** Previous work has investigated the causal role of a single adiposity measure (body mass index; BMI) on metabolites. Using multiple adiposity measures to identify adiposity-metabolite **profiles may reduce bias** and **lead to meaningful interpretations**.

1. **How do we define a profile?**
2. **What makes profiles similar?**

METHODS

We used **Mendelian randomization**, a method that uses **genetic variants** as **proxy** measures for **exposures** to avoid limitations of observational methods, to investigate three measures of adiposity: **BMI**, **WHR** (waist hip ratio) and **BF** (body fat percentage). **Variants** are inherited at birth and **stable over time** allowing for **causal interpretation**. We used genetic data for our exposures from up-to 795,624 Europeans and investigated associations with **123 NMR metabolites** from 24,925 Europeans. We used an inverse variance weighted model and **grouped metabolites** using subclasses to **define profiles.**

1. **How can we quantify** dis/similarity of **profiles**
2. **How** else **can we group metabolites?**



Circos plot: Mendelian randomization analysis results using an **inverse variance weighted** (multiplicative Random effects) model of three exposures (**body mass index** (BMI), **waist hip ratio** (WHR) and **body fat percentage** (BF)) and 123 NMR derived **metabolites**. Units; 1-SD increase in exposure per SD increase in outcome; 95% confidence interval.

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Scan for: digital poster, more info, data sources, sensitivity analyses and pretty pictures
mattlee821.github.io/002_adiposity_metabolites